

Shivsai V Sadare

Bengaluru, India — +91 7892300900 — sadareshivsaiv@gmail.com
linkedin.com/in/shivsai-v-sadare-1a35202ba — github.com/Shivsai14

EDUCATION

BNM Institute of Technology

Bachelor of Engineering in Electronics and Communication Engineering
CGPA: 6.89/10.0 — Currently in 7th Semester

Aug 2023 – May 2027 (Expected)

Bengaluru, India

SGPTA PU College

2nd PU (12th Standard), State Board

Graduated 2023

64.83%

Sri Chaitanya Techno School

10th Standard, CBSE

Graduated 2021

82.60%

TECHNICAL SKILLS

Programming Languages: C, Python, Java, System Verilog, Data Structures & Algorithms (DSA)

Technologies & Frameworks: Machine Learning (ML), Deep Learning, AI, IoT, VLSI, Analog IC Design

Hardware & Embedded Systems: HTML/CSS, Arduino, ESP-32, Raspberry Pi, Digital IC Design

Tools: Git, GitHub, Cadence Virtuoso, Microsoft Office Suite, Simulation Software

Areas of Expertise: Software Development, Embedded Systems, Signal Processing, Web Development

PROJECTS

Project Aegis: AI-Powered Predictive Maintenance System

April 2026

- Engineered an autonomous industrial monitoring agent using Python and Gemini 1.5 Flash to monitor live sensor telemetry for four machines simultaneously.
- Implemented confidence-gated diagnostic logic to trigger maintenance alerts only when diagnostic certainty exceeded 80%, minimizing false positives.
- Developed a real-time "Mission Control" dashboard using React and FastAPI to visualize risk scores, remaining useful life (RUL), and autonomous recovery roadmaps.

Agentic AI for STEM Education

March 2025

- Developed an AI-powered personalized learning platform to enhance STEM education engagement using Python.
- Implemented intelligent agent-based systems to adapt content delivery based on student learning patterns.

Climate Change Control using Satellite Images

February 2025

- Built a computer vision system to analyze satellite imagery for environmental change detection.
- Processed and classified large-scale geospatial data to identify climate-related patterns and anomalies.

Audio Signal Compression using Finite State Machine

January 2024

- Designed and implemented FSM-based compression algorithm for efficient audio data processing.
- Optimized signal quality retention while achieving significant compression ratios.

EXPERIENCE

System Verilog Intern — Cranes Varsity

2024

- Gained hands-on experience in Hardware Description Language (HDL) and digital design verification.
- Developed testbenches and verification environments for digital circuits using System Verilog.

ACHIEVEMENTS & CERTIFICATIONS

- VLSI SoC Design using Verilog HDL** – Maven Silicon (May 2026).
- Analog IC Design Implementation and Verification** – Cadence Training Services.
- Java Programming Fundamentals** – Infosys Springboard.
- IBM Hackathon Finalist** – Selected among top teams for innovation and algorithmic problem-solving.
- Semiconductor Fabrication Technologies Certification** – Indian Institute of Science (IISc).
- Deep Learning and Applications FDP** – Successfully completed Faculty Development Program.
- Digital Productivity with AI Certification** – YuWaah! Platform.
- Network Addressing & Troubleshooting Certification** – Cisco Networking Academy.

ADDITIONAL INFORMATION

Languages: English (Fluent), Hindi (Fluent), Kannada (Native), Marathi (Proficient)

Soft Skills: Communication, Time Management, Team Collaboration, Problem-solving, Adaptability

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.